Stat 220: Probability as a Model

Midterm practice questions, with answers

Practice on choosing a distribution, fitting it by likelihood, checking the fit, and turning a prediction interval back into the distribution it came from. Roughly half of these want a number, and the arithmetic is small enough to do in the margin. No computer needed.

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1. A moving company sends 300 quotes in a month and wants to model how many become bookings. Which distribution fits the way that outcome arises?

   A. Poisson, because bookings are a count.

   B. Binomial, with $n = 300$ and p the booking probability.

   C. Normal, because 300 is a large sample.

   D. Exponential, because bookings arrive over time.

   Answer: B. There is a *fixed* number of tries, 300, and each one either converts or does not. That is the binomial story. A Poisson would be right for bookings per hour, where there is no fixed ceiling on the count. The sample size being large says something about the sampling distribution of $\hat{p}$, not about the distribution of the count itself.

2. Flights of stairs per moving job have mean 1.17 and variance 1.18. Estimated boxes per job have mean 59 and variance 399. What do these four numbers tell you before you plot anything?

   A. Both are consistent with a Poisson model; the boxes just have a bigger mean.

   B. Stairs are consistent with a Poisson; boxes are not, because the variance is far above the mean.

   C. Neither is Poisson, because a Poisson needs a mean below 1.

   D. Boxes are the better Poisson candidate, because larger counts are closer to Poisson.

   Answer: B. A Poisson has one spread available to it, equal to its mean. Stairs come in at a ratio of 1.01 and boxes at 6.8, so the boxes carry almost seven times the variation the model allows. This is the cheapest check in the unit and it needs no figure.

3. Why is a histogram on its own a weak basis for choosing a distribution?

   A. Different processes can produce similar-looking histograms, and with a small sample the shape is hard to judge at all.

   B. The shape depends on the bin width, so no shape is reliable.

   C. Histograms are only valid for continuous data.

   D. A histogram shows density rather than probability.

Answer: A. The bin width is a real annoyance but not the main point. The main point is that the histogram shows you how the data look and tells you nothing about how they arose, and the mechanism is what picks the distribution. Forty observations can look like almost anything.

4. A model assumes a constant failure rate, so that after 30 hours without a breakdown the chance of surviving the next 10 hours is the same as it was at the start. Which model is this, and when is that assumption a problem?

   A. Poisson; a problem when events are not independent.

   B. Normal; a problem when the data are skewed.

   C. Exponential; a problem when parts wear out, because the hazard then rises with age.

   D. Binomial; a problem when p changes between tries.

   Answer: C. Memorylessness is the exponential's signature and its biggest liability. A truck transmission is more likely to fail in its tenth year than its first, and an exponential model cannot represent that: it assigns the same conditional risk forever.

5. Of 600 moving jobs, 163 ran on a weekend, and $\mathcal{L}(p) = p^{163}(1-p)^{437}$. What is $\mathcal{L}(0.20)$?

   A. The probability that p equals 0.20.

   B. The probability of seeing exactly this pattern of weekend and weekday jobs if p were 0.20.

   C. The probability that p is less than 0.20.

   D. The proportion of jobs that would run on a weekend if p were 0.20.

   Answer: B. A likelihood is a probability of the *data*, computed at a candidate parameter value. It is not a probability distribution for p: the curve does not integrate to 1 over p, and asking for "the probability that $p = 0.20$" is a Bayesian question that needs a prior, which is Unit 10.

6. Maximizing $p^{163}(1-p)^{437}$ gives $\hat{p} = 163/600$. What is the point of going through likelihood to reach a number you would have written down anyway?

   A. Likelihood gives a different and better answer in large samples.

   B. The sample proportion is only an approximation; likelihood gives the exact value.

   C. The method agrees with common sense in the easy cases and keeps working in the hard ones, like logistic and Poisson regression, where no obvious estimator exists.

   D. It confirms that the sample proportion is unbiased.

   Answer: C. This is the whole argument for the machinery. You check a general method against cases where you already know the answer, and then you can trust it where you do not. Logistic regression has no closed-form estimator and is fitted by exactly this search.

7. Software maximizes $\log \mathcal{L}$ rather than $\mathcal{L}$ itself. Why?

   A. The log-likelihood has a different maximum, which is the one you want.

   B. Taking the log is required before you can use calculus.

   C. The log turns the product into a sum, which avoids the underflow a product of thousands of small probabilities runs into, and it peaks in the same place.

   D. The log-likelihood is always positive, which makes it easier to maximize.

   Answer: C. At 600 observations the likelihood is already around 4×10^{-153}. A few thousand more and it is exactly zero in floating point, leaving the optimizer a flat surface with nothing to climb. The log is a sum, it never underflows, and because log is increasing it has its maximum at the same parameter value. Note that a log-likelihood is usually negative.

8. Two likelihood curves for a proportion have the same peak location but one is much broader. What does the broader curve mean?

   A. A biased estimate.

   B. More uncertainty about the parameter: nearby values fit almost as well.

   C. A larger sample size.

   D. A parameter further from 0.5.

   Answer: B. Width is uncertainty. A broad peak says the data cannot distinguish $\hat{p}$ from its neighbours, which is exactly what a large standard error says. Formally, the curvature of the log-likelihood at the peak gives an approximate SE, so flatter means larger. Smaller samples give broader curves, not larger ones.

9. AIC $= -2 \log \mathcal{L} + 2k$. A colleague compares a model for 'price' against a model for 'log(price)' by their reported AICs and picks the lower one. What is wrong?

   A. Nothing, as long as both models use the same predictors.

   B. AIC cannot be used with logged variables at all.

   C. AIC is only valid for comparing models with the same number of parameters.

   D. The two likelihoods describe different outcomes, so the AICs are not on a common scale without an adjustment for the change of variable.

   Answer: D. AIC compares fits to the *same* observations on the *same* scale. A density for $\log y$ is not a density for y, and converting between them brings in a Jacobian term. Without that correction the comparison is meaningless, and it usually flatters the logged model. The $2k$ term already handles differing parameter counts, which is its job.

10. A Poisson is fitted to the age in years of 500 delivery vans. The variance-to-mean ratio comes out 1.16, so the cheap check passes. The observed-against-expected table then shows no vans

aged 0 where the model expects 5, and 65 vans aged 8 where it expects 24. What is the lesson?

A. A check on the feature you fitted can barely fail; only a comparison the model was not fitted to can tell you whether the shape is right.

B. The sample is too small for either check to mean much.

C. The ratio check is unreliable and should not be used.

D. The model is fine, since 1.16 is close enough to 1.

Answer: A. The ratio is one summary, and a column that is roughly uniform on 1 to 8 happens to land near a ratio of 1 by coincidence. The table looks at the whole shape and rejects the model flatly. Keep the ratio check: it is free and it catches overdispersion. Just do not stop there.

11. A goodness-of-fit test on 50,000 observations returns $p = 0.001$ for a fitted Poisson, and the largest observed-against-expected gap anywhere in the table is 2%. What should you report?

A. The model is rejected, so it should not be used.

B. The test is invalid because the sample is too large.

C. The departure is real but small; whether it matters depends on which part of the distribution your question uses.

D. The p-value should be adjusted for the number of cells.

Answer: C. Large samples make small departures significant, which is the test doing its job rather than misbehaving. A 2% mismatch in the middle is irrelevant to an estimate of the mean and could be fatal to a 99th-percentile capacity plan. Report the size of the mismatch alongside the p-value and say where in the distribution your answer lives.

12. A normal Q-Q plot of 700 job times is straight through the middle and curls upward at both ends. What does that pattern indicate?

A. Lighter tails than normal, suggesting a floor and a ceiling.

B. Heavier tails than normal in both directions.

C. Repeated values or rounding.

D. A right-skewed distribution with a long upper tail.

Answer: D. Both ends curling *up* is the right-skew signature: the large values are larger than normal quantiles predict and the small values are not as small. Both ends bending *away* from the line, one down and one up, is the heavy-tailed pattern. A flat stretch of points means ties.

13. A normal model fitted to 700 lawn mowing jobs has mean 15.3 minutes and SD 6.7. Which single finding is the strongest argument against it?

A. The observed share within one SD is 76.7% rather than 68%.

B. Skewness is 1.94 rather than 0.

C. The model says 1.4% of jobs run past 30 minutes when 3.4% do.

D. The model gives a 1.1% chance of a job taking less than zero minutes.

Answer: D. All four are evidence and the third is of a different kind. The others say the model is inaccurate; a 1-in-90 chance of finishing before you start says the model is *about the wrong kind of quantity*. A positive, right-skewed duration wants a lognormal, and on these data logging the outcome takes the skewness from 1.94 to 0.04.

14. A density curve for job hours reaches a height of 1.4 at $x = 3$. What does that tell you?

A. Nothing on its own: density is height, probability is area, and a height may exceed 1.

B. About 14% of jobs take close to 3 hours.

C. There is a 140% chance of a value near 3, which is impossible, so the curve is wrong.

D. Three hours is the most likely single value.

Answer: A. Heights are not probabilities and are not capped at 1. The total *area* is 1, so a narrow distribution has to be tall: a normal with SD 0.2 peaks just under 2.0. To get a probability you need an interval.

15. For a continuous outcome X, how do $\mathbb{P}(X \geq 16)$ and $\mathbb{P}(X > 16)$ compare?

A. They differ by the density at 16.

B. They are equal, because a single point contributes zero area.

C. The first is larger, by the probability of exactly 16.

D. It depends on whether the distribution is skewed.

Answer: B. $\mathbb{P}(X = 16) = 0$ for any continuous distribution, so the two are the same number. This stops being true the moment you round: if times are recorded to the nearest tenth of an hour, "16.0" stands for an interval and carries positive probability.

16. A value sits 2.5 standard deviations above the mean. A colleague concludes it is in the top 0.6% of the distribution. When is that right?

A. Always; that is what a z-score means.

B. Only when the sample size is above 30.

C. Only when the data are continuous.

D. Only when the distribution is normal, since the 0.6% comes from the normal curve rather than from the z-score.

Answer: D. A z-score is a unit of distance and carries no information about shape. The map from z to a tail area is the normal distribution's, and in a right-skewed distribution $z = 2.5$ can have several times that tail probability. The same mistake sits behind quoting 68-95-99.7 for data that are not normal.

17. A logged model predicts $\widehat{\log(\text{cost})} = 6.2$, with residual SD 0.6 on the log scale. $e^{6.2} = 493$. Which is the median predicted cost and which is the mean?

    A. Median 493; mean $493 \times e^{0.6^2/2} \approx 590$.

    B. Mean 493; median $493 \times e^{0.6^2/2} \approx 590$.

    C. Both are 493, since exponentiating is just a change of units.

    D. Median 493; mean $493 \times e^{0.6} \approx 898$.

    Answer: A. Exponentiating preserves order, so it carries the middle value to the middle value: $e^{\hat{y}}$ is the median. The mean is larger by $e^{\sigma^2/2} = e^{0.18} = 1.20$, so about \$590. Quote 493 to one customer and budget 590 a job across many. The factor is $e^{\sigma^2/2}$, not e^{σ}.

18. On Unit 5's diamond model the residual SD on the log scale is 0.13, so the mean sits only 0.9% above the median. What should you conclude about the mean-median distinction?

    A. It is a technicality that can be ignored in practice.

    B. It matters only for prices.

    C. The gap grows with the residual spread, so it is negligible here and reaches 38% at $\sigma = 0.8$; check σ before deciding.

    D. It applies only when the sample size is small.

    Answer: C. The honest answer is that the rule is real and its size is an empirical question. $e^{\sigma^2/2}$ is 1.009 at $\sigma = 0.13$ and 1.377 at $\sigma = 0.8$. A budget built on the median would be 38% short in the second case and rounding error in the first.

19. A 95% prediction interval for one delivery route runs from 40 to 70 minutes. Treating the forecast as normal, what is SE_{pred}?

    A. $70 - 40 = 30$ minutes.

    B. $(70 - 40)/(2 \times 1.96) \approx 7.7$ minutes.

    C. $(70 - 40)/2 = 15$ minutes.

    D. $(70 - 40)/\sqrt{2} \approx 21$ minutes.

    Answer: B. A 95% interval is $\hat{y} \pm 1.96\,\text{SE}_{\text{pred}}$, so the full width is $2 \times 1.96 = 3.92$ standard errors. Dividing by 3.92 gives 7.65. Once you have that, the forecast is Normal(55, 7.65^2) and any threshold question is one z-score away.

20. For one moving job the confidence interval for the average such job is 12.25 to 12.54 hours, and the prediction interval for this one job is 10.74 to 14.04. Collecting ten times as much data would do what?

    A. Shrink both intervals by about a factor of three.

    B. Shrink the prediction interval and leave the confidence interval unchanged.

    C. Shrink the confidence interval and leave the prediction interval nearly unchanged.

    D. Leave both unchanged, since they describe the population.

    Answer: C. $\mathrm{SE}_{\text{pred}}^2 = \mathrm{SE}_{\text{line}}^2 + \sigma^2$, which here is $0.073^2 + 0.836^2$. More data drives the first term toward zero and does nothing to the second: jobs with identical predictors still differ from each other. A prediction interval has a floor and the floor is the residual spread.

21. A job's time is forecast as Normal(12.4, 0.84^2) hours. Three such jobs run back to back and independently. What is the SD of the total?

    A. $0.84 \times \sqrt{3} = 1.45$ hours.

    B. 0.84 hours, unchanged.

    C. $0.84 \times 3 = 2.52$ hours.

    D. $0.84/\sqrt{3} = 0.48$ hours.

    Answer: A. Variances add, so the total's variance is 3×0.84^2 and its SD is $0.84\sqrt{3} = 1.45$. Adding the SDs to 2.52 is the common error and it is not small: $\mathbb{P}(\text{total} > 40)$ comes out 2.6% the right way and 13% the wrong way, a factor of five. Dividing by $\sqrt{3}$ is the formula for the SD of an *average*, not a total.

22. A crew earns overtime past 13 hours at \$85 an hour. The point prediction for a job is 12.4 hours. What is the expected overtime cost?

    A. Positive, because $\mathbb{E}[g(Y)] \neq g(\mathbb{E}[Y])$ for a rule with a kink; the right answer needs the whole distribution and comes to about \$9.80.

    B. \$0, since 12.4 is below 13.

    C. $85 \times (13 - 12.4) = \$51$.

    D. It cannot be computed without knowing the actual time.

    Answer: A. Plugging the average into a nonlinear cost rule throws away the cases that cost money. The point prediction says no overtime; the distribution says 23% of such jobs run past 13 hours, and averaging the cost over the forecast gives $\mathbb{E}[(Y-13)^+] = 0.115$ hours, about \$9.77. Across a thousand bookings that is \$9,800 the point prediction never showed you.

23. A regression on moving jobs with volumes from 120 to 1,317 cubic feet is asked to predict a 2,000 cubic foot job. The software returns 27.3 hours with a prediction interval of 25.7 to 29.0.

What is the right reaction?

A. Use it; the interval is narrow, so the prediction is precise.

B. Widen the interval by the ratio 2000/1317.

C. Refit without the large jobs and try again.

D. The interval describes uncertainty *within* the model, and nothing in the output signals that there is no data out there to support the model at all.

Answer: D. This is the failure mode with no warning attached. Every number in that interval is computed as though the fitted relationship still holds at 2,000 cubic feet, and no column of the output can tell you whether it does. The narrowness is a property of having 600 observations, not of having relevant ones.

24. You need the 99th percentile of job times for a capacity plan. Your fitted normal model matches the data well in the middle and runs a little light in the upper tail. What follows?

A. The model is adequate, since it fits where most of the data are.

B. Report the 99th percentile with a note that the model fits well overall.

C. Use the sample's own 99th percentile instead, which is always preferable.

D. The one region the answer depends on is the one region the model gets wrong, so the plan would be under-resourced; check or model that tail directly.

Answer: D. Always ask which part of the distribution the answer lives in. A model that is excellent in the middle is irrelevant here, and a light tail means the capacity estimate is biased in the dangerous direction. The empirical percentile is one option but it rests on very few observations, which is its own problem.